

ENCI335

Structural Analysis and Systems I

Tutorial 4 Solution

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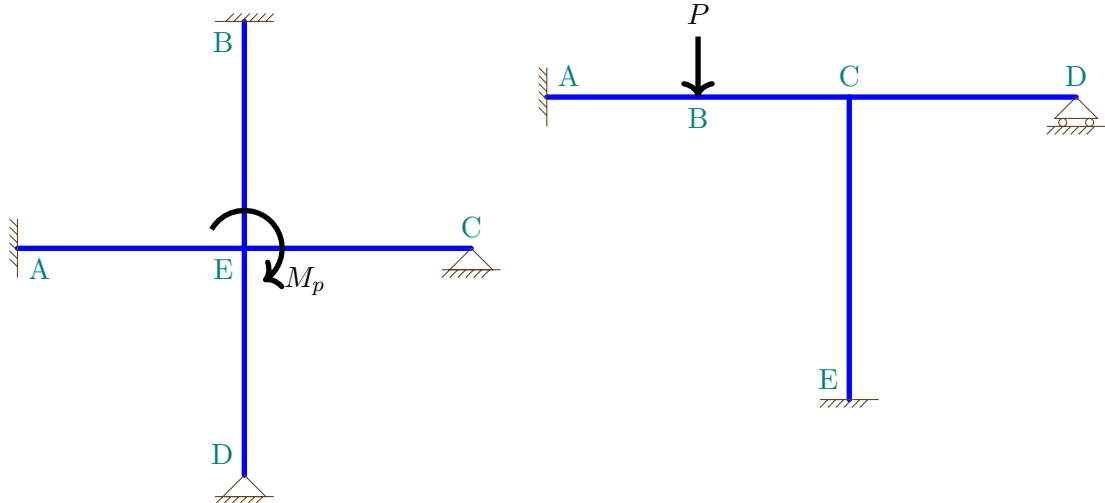
Objectives

This tutorial provides an opportunity to practice the following skills:

- Analysis of statically indeterminate frames using Moment-Area Theorems.
- Sketch the BMD, SFD, AFD, and deflected shape of frames.

Question 1 Deflected Shapes, BMD and SFD

Sketch qualitatively the deflected shapes, BMD and SFD of the following frames, and report their SI .

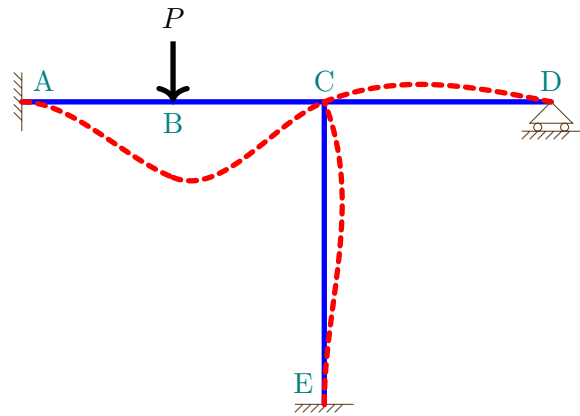
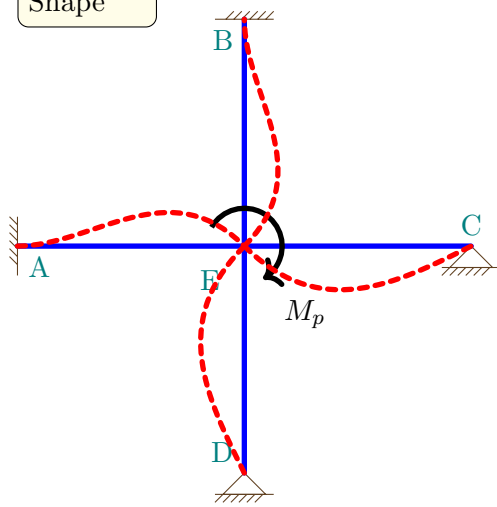


Solution:

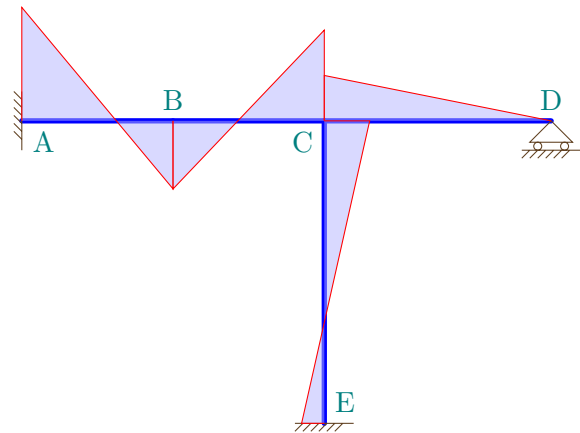
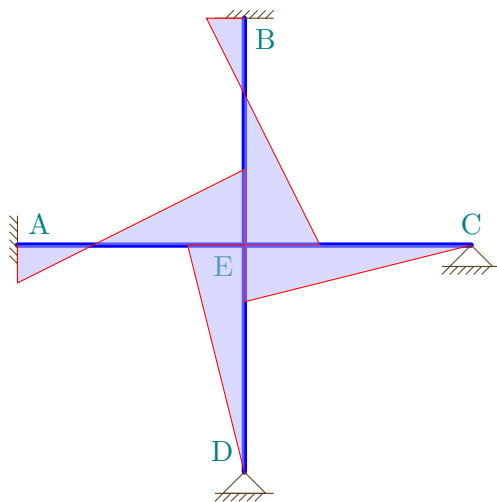
$$\text{Frame 1: } n_R = 10, \quad n_M = 4, \quad n_J = 5, \quad SI = n_R + 3(n_M - n_J) = 7 \quad (1)$$

$$\text{Frame 2: } n_R = 7, \quad n_M = 3, \quad n_J = 4, \quad SI = n_R + 3(n_M - n_J) = 4 \quad (2)$$

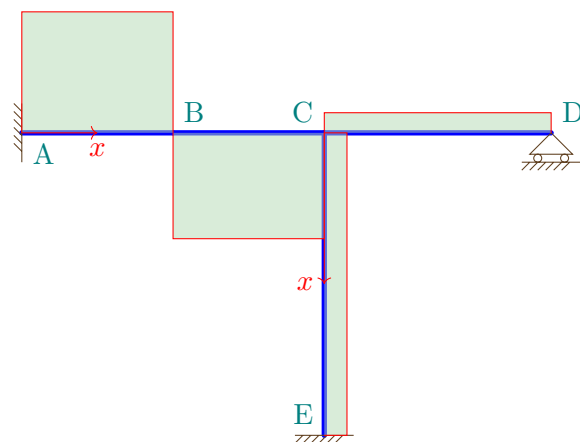
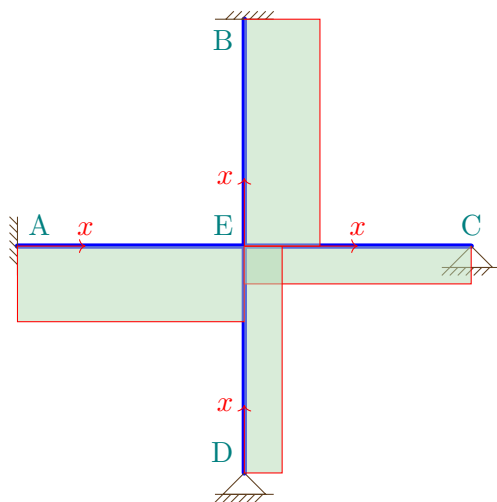
Deflected Shape



BMD

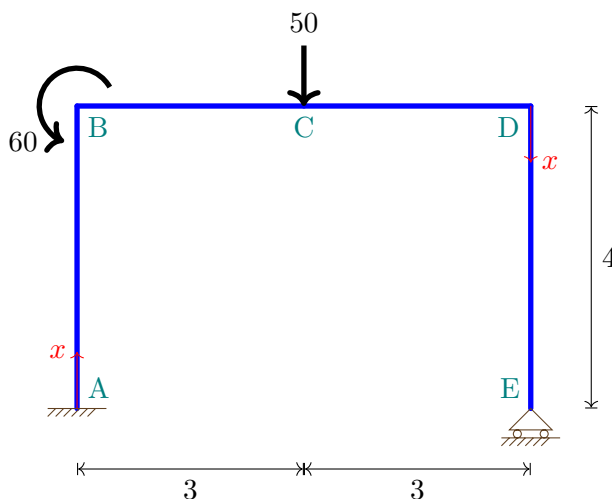


SFD



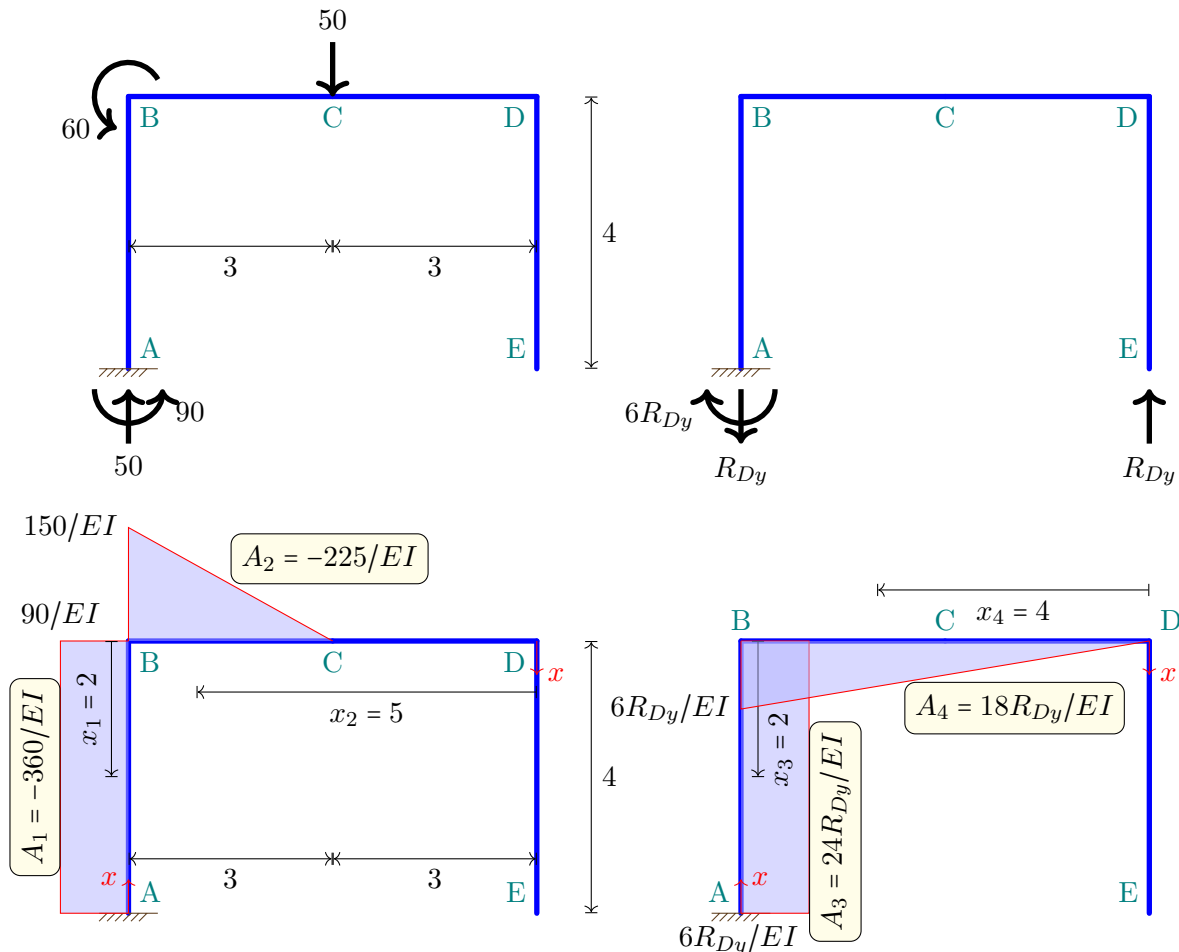
Question 2 Moment-Area Theorems

Solve and sketch the BMD, SFD, AFD, and deflected shape of the following frame (with a uniform flexural rigidity EI) using the Moment-Area theorems. Confirm that the horizontal deflection at E is $205/EI$ ($\rightarrow$).



Solution:

Given a fixed support and a roller support, there are 4 unknown reactions. There is no internal hinge, the frame is, therefore, statically indeterminate to the first degree. We use the vertical support reaction at point E as the unknown redundant, thereby creating two load cases as follows, with their support reactions computed using the equilibrium equations and their respective M/EI diagram.



The vertical deflection at E, v_E is equivalent to the vertical deflection at D, v_D , which can be calculated as

$$v_D = v_B + \theta_B L_{BD} + \delta_{DB} \quad (3)$$

Since the support at A is zero, $v_B = 0$. Since θ_A is zero, the rotation $\theta_B = \theta_{BA} = A_1 + A_3$. Since the support at E is zero, $v_D = 0$. Therefore, we have

$$0 = (A_1 + A_3)6 + (A_2 \cdot x_2 + A_4 \cdot x_4) \quad (4)$$

Upon the substitution of the values of A 's and x 's into the above equation, we obtain that $R_{Dy} = 15.21$.

The reactions at A are $R_{Ay} = 50 - R_{Dy} = 34.8$, and $M_A = 90 - 6R_{Dy} = -1.25$ (clockwise).

To find the horizontal deflection at E, we first find the horizontal deflection at B is

$$u_B = u_A + s_{AB} \cdot (\theta_A L_{AB} + \delta_{BA}) \quad (5)$$

Given $u_A = 0$ and $\theta_A = 0$, and $s_{AB} = -1$, therefore

$$u_B = -(A_1 \cdot x_1 + A_3 \cdot x_3) = \frac{-10}{EI} \quad (\leftarrow) \quad (6)$$

Given that beam BCD is axially rigid, therefore

$$u_D = u_B = \frac{-10}{EI} \quad (\leftarrow) \quad (7)$$

The rotation at B is

$$\theta_B = A_1 + A_3 = \frac{5}{EI} \quad (8)$$

The rotation at D is

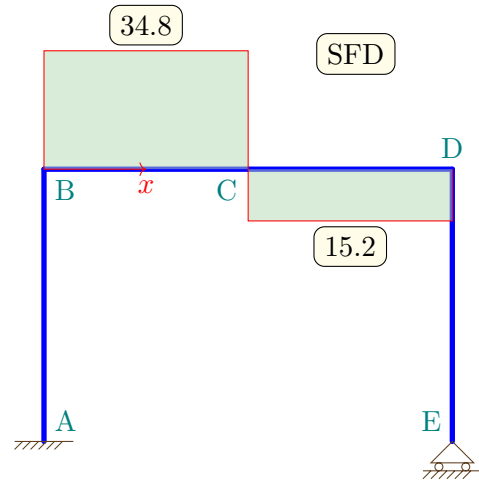
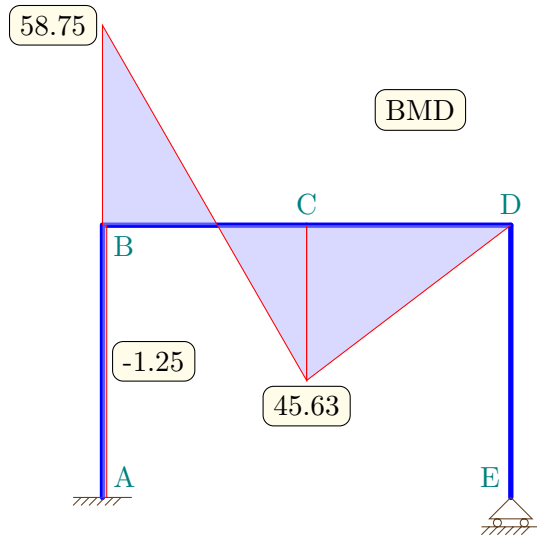
$$\theta_D = \theta_B + A_2 + A_4 = A_1 + A_3 + A_2 + A_4 = \frac{53.75}{EI} \quad (9)$$

The horizontal deflection at E, with $s_{DE} = 1$, is

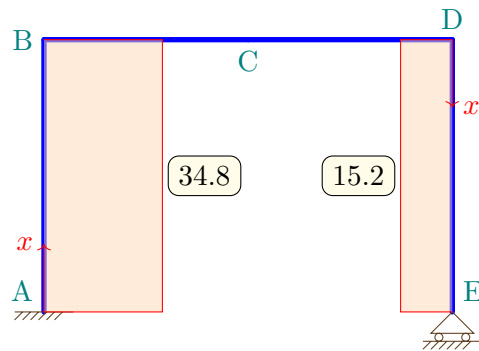
$$u_E = u_D + s_{DE} \cdot (\theta_D L_{DE} + \delta_{ED}) \quad (10)$$

$$= \frac{-10}{EI} + \frac{53.75}{EI} \cdot 4, \quad (\delta_{ED} = 0) \quad (11)$$

$$= \frac{205}{EI} \quad (\rightarrow) \quad (12)$$



AFD



Deflected Shape

